The Physics Notebook

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1 Prerequisite Mathematics

§1.1 Vectors

Vectors are quantities with magnitude and direction, contrastingly to scalars which only have magnitude. For our purposes, vectors can be understood as directed line segments, although they're more rigorously defined as elements of a vector space, or can be understood through rotations. We'll talk about rotations later, but will not deal with vector spaces right now.

I shall use boldface to denote vectors, such as $\mathbf{A}$.

The utility of vectors stem from their independence from a particular co-ordinate system. They're directed *line segments*, only their length matters and we're not concerned with their end points. This means that a vector can be freely translated.

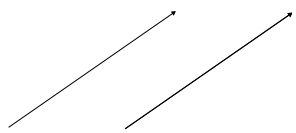


Figure 1.1: Identical Vectors

The magnitude of a vector, its norm, is its euclidean "length", we denote it by $\|\mathbf{A}\|$ or simply A.

A unit vector, $\hat{\mathbf{a}}$ ("a hat") is a vector whose magnitude is unity. We use the unit vector to often denote the direction of a vector by multiplying the unit vector with its magnitude. For instance, the unit vector that point is the direction of A, $\hat{\mathbf{A}}$ can be calculated as,

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{A}$$

where we divide by the magnitude of A to ensure that the magnitude of $\hat{\mathbf{a}}$ is 1.

§1.2 Vector Algebra

There are two operations generally defined on vectors, *scalar multiplication* and *vector addition*.

§1.2.1 Scalar Multiplication

Multiplying a vector $\mathbf{A}$ by a scalar c results in another vector, $c\mathbf{A}$ parallel or antiparallel(pointing in the opposite direction) to $\mathbf{A}$ scaled by a factor of $|c|$.

If c is positive, then the resultant vector is simply scaled by it. If we multiply a vector by -1, then the result is the simple reversal in direction of vectors. If c is negative, then the vector is reversed and scaled by $|c|$.

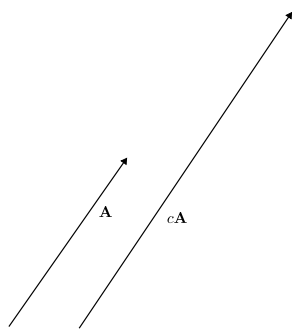


Figure 1.2: Scalar Multiplication

§1.2.2 Vector Addition

We define an operation called addition of vectors which produces another vector. To add two vectors, $\mathbf{A}$ and $\mathbf{B}$, we stick the tail of one vector with the head of another.

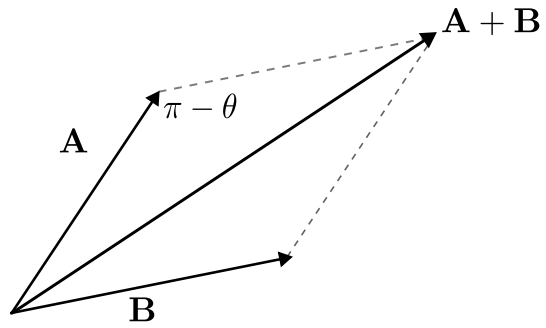


Figure 1.3: Vector Addition

We can calculate the magnitude of their resultant vector, $\|\mathbf{A}+\mathbf{B}\|$ by using the cosine rule. In §1.2.2 θ is the angle between $\mathbf{A}$ and $\mathbf{B}$.

$$\begin{aligned}
\|\mathbf{A}+\mathbf{B}\|^2 &= A^2+B^2-2AB\cos(\pi-\theta)\\
\|\mathbf{A}+\mathbf{B}\|^2 &= A^2+B^2+2AB\cos\theta\\
\|\mathbf{A}+\mathbf{B}\| &= \sqrt{A^2+B^2+2AB\cos\theta}
\end{aligned}$$

To subtract two vectors, we simply reverse the vector and then add the resulting vector.

Example 1.2.1

Consider the infinitesimal change vector, $\mathrm{d}\mathbf{A}=\mathbf{A}(t+\mathrm{d}t)-\mathbf{A}(t)$.

We have,

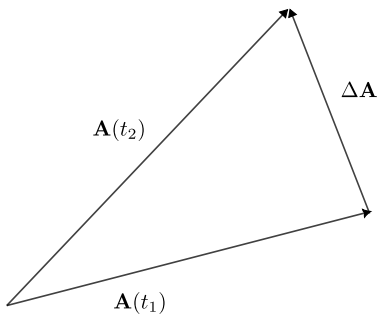
$$\begin{aligned}
\mathrm{d}\mathbf{A} &= A(t+\mathrm{d}t)\hat{\mathbf{A}}(t+\mathrm{d}t)-A(t)\hat{\mathbf{A}}(t)\\
\mathrm{d}\mathbf{A} &= A(t+\mathrm{d}t)\hat{\mathbf{A}}(t+\mathrm{d}t)-A(t+\mathrm{d}t)\hat{\mathbf{A}}(t)\\
&\quad + A(t+\mathrm{d}t)\hat{\mathbf{A}}(t)-A(t)\hat{\mathbf{A}}(t)\\
\mathrm{d}\mathbf{A} &= A(t+\mathrm{d}t)(\hat{\mathbf{A}}(t+\mathrm{d}t)-\hat{\mathbf{A}}(t))+\hat{\mathbf{A}}(t)(A(t+\mathrm{d}t)-A(t))\\
\mathrm{d}\mathbf{A} &= A\,\mathrm{d}\hat{\mathbf{A}}+\mathrm{d}A\,\hat{\mathbf{A}}
\end{aligned}$$

The first term represents a change in direction and the second represents a change in magnitude.

The magnitude of the first can be calculated as,

Let us first consider a geometric argument for $A\,\mathrm{d}\hat{\mathbf{A}}$ which is perpendicular to $\mathbf{A}$. In the limiting case, $A\,\mathrm{d}\hat{\mathbf{A}}$, the change in magnitude is negligible, we only observe a change in direction. For now, let us consider $\Delta\mathbf{A}$ perpendicular to $\mathbf{A}$ as an approximation.

Consider the following figure,



This allows us to consider a simple geometric argument which gives $\|\Delta\mathbf{A}\| = 2A\sin(\theta/2)$ where θ is the angle between $\mathbf{A}(t_1)$ and $\mathbf{A}(t_2)$.

Since we're considering values of small θ, we may use the small angle approximation, $\sin(\theta) \approx \theta$. Therefore,

$$\|\boldsymbol{\Delta}\mathbf{A}\| \approx 2A \times \frac{\theta}{2} \tag{1.1}$$

$$\approx A \times \theta \tag{1.2}$$

Which is equivalent to $A\Delta\theta$. In the limit $\Delta\theta \to 0$,

$$\mathrm{d}A_\perp = A\,\mathrm{d}\theta \tag{1.3}$$

Which finally gives us,

$$\left\|\frac{\mathrm{d}\mathbf{A}_\perp}{\mathrm{d}t}\right\| = A\frac{\mathrm{d}\theta}{\mathrm{d}t} \tag{1.4}$$

which is the magnitude of $A\,\mathrm{d}\hat{\mathbf{A}}$. Thus,

$$\|\mathrm{d}\hat{\mathbf{A}}\| = \frac{\mathrm{d}\theta}{\mathrm{d}t} \tag{1.5}$$

§1.3 Dot Product

The *dot product* of two vectors results in a scalar and is defined as,

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta$$

where θ is the angle between $\mathbf{A}$ and $\mathbf{B}$.

Note that the quantity $\mathbf{A}\cdot\mathbf{B}$ is just A times the magnitude of the projection of $\mathbf{B}$ on $\mathbf{A}$. Similarly, $\mathbf{B}\cdot\mathbf{A}$ is magnitude of the projection of $\mathbf{A}$ on $\mathbf{B}$ times B.

We may note that,

$$\mathbf{A}\cdot\mathbf{A} = A^2$$

Or,

$$A = \sqrt{\mathbf{A}\cdot\mathbf{A}}$$

It can be easily shown that the scalar product distributes over vector addition. Another thing of importance is the identity(which is also easy to derive),

$$\frac{\mathrm{d}\mathbf{A}\cdot\mathbf{B}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{A}}{\mathrm{d}t}\cdot\mathbf{B} + \frac{\mathrm{d}\mathbf{B}}{\mathrm{d}t}\cdot\mathbf{A}$$

§1.4 Cross Product

The *cross product* is another product operation on vectors, this product produces a vector.

It is defined as,

$$\mathbf{A}\times\mathbf{B} = AB\sin\theta\hat{\mathbf{n}}$$

Where $\hat{\mathbf{n}}$ is the vector perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$. Since there are two directions that $\hat{\mathbf{n}}$ can have, we define the two vectors and their cross product to form a right-hand triple.

Place your fingers in the direction of $\mathbf{A}$ and curl them, along the smaller angle, towards $\mathbf{B}$. Then the direction in which your thumb is pointing is the direction of $\mathbf{A}\times\mathbf{B}$.

§1.5 Vectors in Component form

Although we have worked without co-ordinate systems till now, to actually extract any meaning from vector operations, a coordinate system is necessary though the concept it describes is actually independent of any coordinate system. A spatial, three-dimensional coordinate system essentially establishes three basis vectors.

Basis vectors are vectors that are linearly independent. That is, their linear combination,

$$c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + \cdots + c_n\mathbf{e}_n$$

only has a trivial solution of $c_1 = c_2 = \cdots = c_n = 0$, when equated to the n dimensional null vector, $c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + \cdots + c_n\mathbf{e}_n = \mathbf{0}$.

2 Unit and Dimensions

Units are dimension are in many ways what give the abstract ideas of physics some ground in the real life. The dimensions of a physical quantity represent what kind of physical quantity is, while the unit is a measure of that physical quantity.

The distinction is appreciable to know, just not always useful.

§2.1 Dimensions

There are seven fundamental quantities which are assigned specific dimensions. It is not necessary to take these as fundamental, in some perfectly fine formulations like the Planck quantities this is not the case. But we'll not be working in these for now.

Quantity	Unit Name	Unit symbol	Dimension
Mass	kilogram	kg	M
Length	meter	m	L
Time	second	s	T
Electric Current	ampere	A	I
Luminous Intensity	candela	cd	J
Temperature	kelvin	K	Θ
Amount of substance	mole	mol	N

Table 2.1: The 7 fundamental quantities.

For any physical quantity A, we use the notation $[A]$ to denote its dimension. For instance,

$$[F] = MLT^{-2}$$

Evaluating the dimensions has a really neat use, it allows us to check our answers, and establish some relations through the use of *dimensional* analysis.

PART A

Mechanics

Part A: Contents

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3 Kinematics

Kinematics is the study of motion without concerning its cause. It allows us to calculate and find out the evolution of a body. Everything in our world undergoes motion, one way or the other. To begin our study of motion, let us first concern ourselves with some definitions.

§3.1 Reference Frame and Point Particle

Motion of any object is considered to relative to a reference frame. We consider motion of a body with respect to any other body or co-ordinate frame.

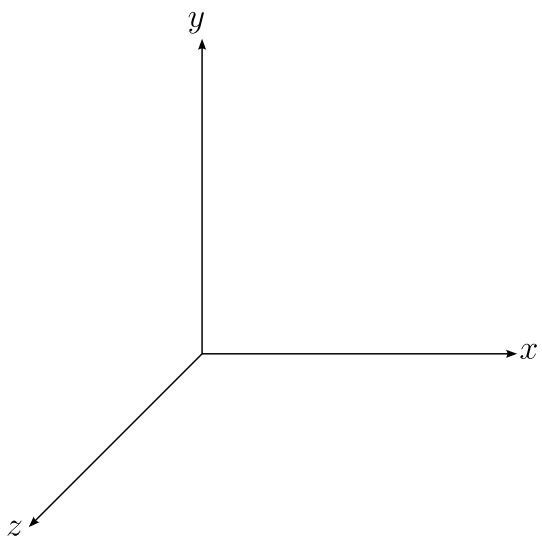


Figure 3.1: A Reference Frame

We use a reference frame according to our convenience. In general, we use the 3-dimensional cartesian system like the one in fig. 3.1.

Definition 3.1.1 (Reference Frame)

A reference frame is a co-ordinate frame relative to which motion of a particle is considered.

We always consider the motion of a point particle. Well, why? It is done so to simplify the calculation of our system. The motion of a (rigid)body is discussed later.

Definition 3.1.2 (Point Particle)

A particle whose size is negligible in the study of its motion is called a point particle.

§3.2 Position Vector and Displacement

§3.2.1 Position Vector

▶ One thing of importance is that a position vector of a particle is 'generally' a function of time. We generally denote this as $r(t)$ to show that is a function.

We define the position of a particle relative to a reference frame using a *position vector*. One end of the position vector is the origin of our reference frame. The other is the particle itself. Consider fig. 3.2, the vector $\mathbf{r}$ from O to A. It describes the position of A relative to the co-ordinate frame. Let $A = (x, y, z)$. Then we denoted $\mathbf{r}$ as $\mathbf{r}(x, y, z)$.

The position to any point is written as,

$$\mathbf{r} = (x, y, z) = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$$

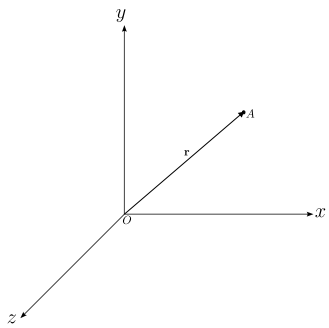


Figure 3.2: A position vector, $\mathbf{r}$.

§3.2.2 Displacement

▶ Note that the position vector is not a *true* vector. It is tied to particular reference frame.

Consider the movement of a particle from $A = (x_1, y_1, z_1)$ to $B = (x_2, y_2, z_2)$. The *displacement* defines a true vector $\mathbf{S} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$. $\mathbf{S}$ is called the displacement vector. Note that it contains no information about the individual points, but only about the relative position of each. Thus, our choice of reference frame, and thus position vector does not matter when we're concerned with displacement . In fig. 3.3, the vector $\mathbf{S}_{AB}$ defines a displacement from A to B.

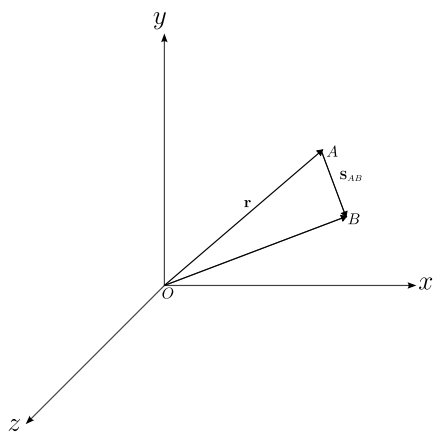


Figure 3.3: Displacement from A to B

The difference between displacement and distance is that of the path they describe. The distance covered is the length of the actual path, while the magnitude of the displacement is simply the length of the vector between the final and initial positions.

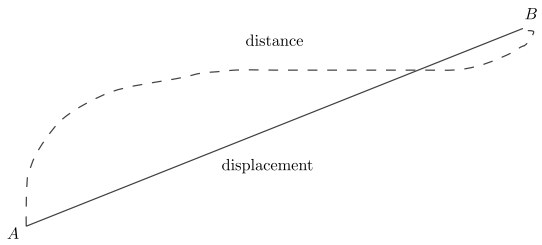


Figure 3.4: Distance vs Displacement

Algorithm 3.2.1 — Consider the case of displacement and distance of a particle in circular motion with radius r Let it move from A to B, inscribing an angle θ (in radians) Then, we have,

Distance between A, $B = r\theta$

Displacement between A, $B = 2r\sin\left(\frac{\theta}{2}\right)$

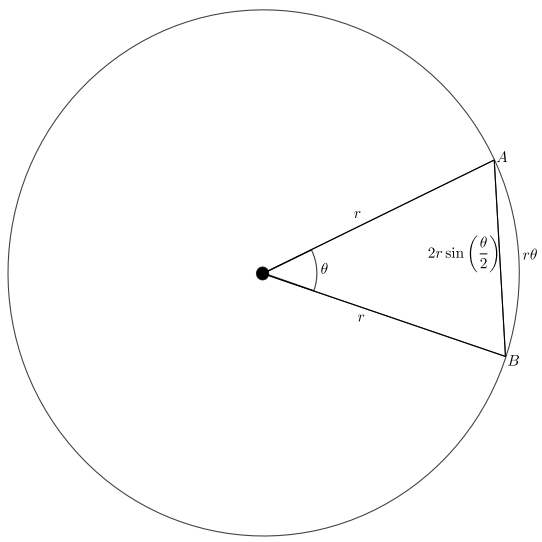


Figure 3.5: Displacement in a circle

§3.3 Speed and Velocity

We are already familiar with the notion of speed. In elementary terms, speed is *distance/time*. Speed itself is an elementary concept. It tells us nothing about the direction of the object whose speed we are talking into consideration. Thus, it is a *scalar* quantity.

The Velocity of a particle, contrastingly, contains information about both the speed and direction of the object in consideration. It is a vector. Similar to position, it is also generally a function of time $v(t)$, though it may also be dependent on the position $v(x)$.

▶ The symbol $\equiv$ stands for 'defined as'.

Definition 3.3.1 (Velocity)

Velocity is defined as,

$$\mathbf{v} \equiv \dot{\mathbf{r}} \tag{3.1}$$

Sometimes, we do not need to compute the derivative of the position vector, and simply need an average estimate, or in the case of uniformly accelerated motion, it is not required that we computer derivatives. In such a case we use the concept of **average velocity**,

$$\langle \mathbf{v} \rangle_{12} = \frac{\mathbf{r}_1 - \mathbf{r}_2}{t_2 - t_1}$$

▶ The dot over r in $\dot{\mathbf{r}}$ tells us that r has been differentiated with respect to time.

Algorithm 3.3.2 — Let us consider the cases of average velocity,

Case I When there are n equal time intervals with velocity $\mathbf{v}_1$, $\mathbf{v}_2$, $\ldots \mathbf{v}_n$. Then,

$$\langle \mathbf{v} \rangle = \frac{\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_n}{n}$$

Case II When there are n equal intervals of distance/displacement with velocity $\mathbf{v}_1$, $\mathbf{v}_2$, $\ldots \mathbf{v}_n$. Then,

$$\langle \mathbf{v} \rangle = \frac{n}{\frac{1}{\mathbf{v}_1} + \frac{1}{\mathbf{v}_2} + \cdots + \frac{1}{\mathbf{v}_n}}$$

In general velocity is also referred to as 'instantaneous velocity'. This is done to remind of the difference with average velocity. We will not refer to it that way since by velocity we shall always mean $\mathbf{v}$ as defined in definition 3.3.1. The same goes for speed. Speed is defined as $\dot{d}$ where d is the distance. Average speed is defined similar to average velocity.

▶ Magnitude refers to the absolute value of a vector, i.e., its modulus. $\|\mathbf{a}\|$ is the magnitude of $\mathbf{a}$ and is often simply denoted a.

There is an interesting relation between speed and velocity. Consider the distance between a and b as $a \to b$. The path between them approaches a straight line. Thus, the distance and displacement become the same in the limiting case. Therefore, the *magnitude* of velocity is the speed at that instant.

Algorithm 3.3.3 — Let us consider some techniques for solving equations of velocity. Consider the case when velocity and time are given and we have to figure out displacement or when displacement and and velocity are given, but time is not.

1. $v = c$: This is a trivial case. Just use equations of uniformly accelerated motion.
2. $v = f(x)$: $\dfrac{\mathrm{d}x}{\mathrm{d}t} = f(x)$. Then, $\dfrac{\mathrm{d}x}{f(x)} = \mathrm{d}t$. Now, we simply have,

$$\int_{x_1}^{x_2} \frac{\mathrm{d}x}{f(x)} = \int_{t_1}^{t_2} \mathrm{d}t$$

   to get x_1 and x_2 or t_1 and t_2. We may also set our co-ordinates such that $x_2 = x$, $x_1 = 0$ and $t_2 = t$, $t_1 = 0$.

3. $v = f(t)$: This is also easy, just, $f(x) = \dfrac{\mathrm{d}x}{\mathrm{d}t}$, then,

$$\int_{t_1}^{t_2} f(t)\,\mathrm{d}t = \int_{x_1}^{x_2} \mathrm{d}x$$

   We simply get x_1 and x_2 or t_1 and t_2 by solving the integral.

§3.4 Acceleration

If a body does not move with uniform velocity, it accelerates. Acceleration is also a vector and is the rate of change of velocity.

Definition 3.4.1 (Acceleration)

Acceleration is defined as,

$$\mathbf{a} \equiv \dot{\mathbf{v}} = \ddot{\mathbf{r}} \tag{3.2}$$

There is also another way to represent acceleration, as,

$$\mathbf{a} = \mathbf{v}\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}x}$$

Such a representation is particularly useful when the velocity of a particle is a function of its position. In such a case, derivating with time is quite a hassle.

§3.5 Uniformly Accelerated Motion

When the acceleration of a particle, a is constant, it undergoes uniformly accelerated motion. We use special cases of integrals of a and v with respect to time and position for deriving the equations of importance.

Theorem 3.5.1

For a particle with initial velocity u undergoing uniformly accelerated motion of acceleration a,

$$v = u + at \tag{3.3}$$

$$v^2 = u^2 + 2as \tag{3.4}$$

$$s = ut + \frac{1}{2}at^2 \tag{3.5}$$

$$s_n = u + \frac{a}{2}(2n - 1) \tag{3.6}$$

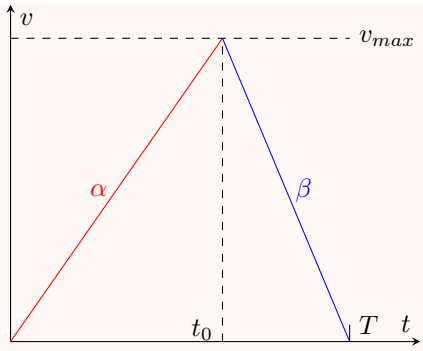
The derivation of these is quite easy, integrate $\int a\,\mathrm{d}x$ for the first, $\int a\,\mathrm{d}t$ for the second (both with constant acceleration), substitute from the former into the latter for the third, and finally for n^{th} second just do $s_n - s_{n-1}$.

Now, we will be looking at some fascinating questions that I have encountered.

Example 3.5.2

Consider a particle moving with acceleration α initially. After some time, it starts to 'de-accelerate' at β acceleration. After time $t = T$, the particle comes to a stop. Find the displacement covered by particle in time T.

Solution. It might be tempting to start calculating after setting up time $t = t_0$ when acceleration changed from α to β, i.e., the particle moves with acceleration α till $t = t_0$. However, let us try to do a cleaner solution. Consider the $v - t$ graph of the particle,



Note that $\alpha(t_0) = \beta(T - t_0) = v_{max}$. Thus,

$$\begin{aligned}
\alpha(t_0) &= \beta(T - t_0) \\
\alpha(t_0) &= \beta(T) - \beta(t_0) \\
\alpha(t_0) + \beta(t_0) &= \beta(T) \\
t_0 &= \frac{\beta T}{\alpha + \beta} \\
v_{max} &= \alpha t_0 \\
v_{max} &= \frac{\alpha\beta T}{\alpha + \beta}
\end{aligned}$$

Now, note that $\frac{1}{2} \times v_{max} \times T$ is the area under the graph.

The area under the $v - t$ is the total displacement.

Thus, displacement, $s = \dfrac{\alpha\beta T^2}{2(\alpha + \beta)}$. □

Example 3.5.3

A car is at distance d from a boy. It starts accelerating at a $\mathrm{m\,s^{-2}}$. What is the minimum velocity that the boy should have to catch up with the car.

Solution. Consider separation of boy and car, $s_{c,b}$. Using the equations of motion, we have $s_b = vt$, $s_c = \frac{1}{2}at^2$. Thus,

$$s_{c,b} = d + \frac{1}{2}at^2 - vt \tag{3.7}$$

The most important thing to note here is that,

For two objects to meet, the solution for time must be real.

Therefore, from the simple equation, (3.7), we must have, for real solutions, $b^2 - 4ac$

for the equation at^2+bt+c. Solving for this by substituting values from (3.7), we have,

$$v \geqslant \sqrt{2ad} \tag{3.8}$$

□

Remark 3.5.4. One thing of note is that the converse of above is true as well,

> **For two objects to *not* meet, the solution of time must be imaginary or negative.**

One more key note to takeaway is that,

At instant of maximum separation, relative velocity of particles is 0.

Example 3.5.5

A body is dropped at $t = 0$, after time $t = t_0$, another body is thrown downwards with velocity u $\mathrm{m\,s^{-1}}$. Assuming first body reaches ground first, plot graph of separation.

Solution. At instant t_0, displacement of first particle $= \frac{1}{2}gt_0^2$. Note that here we set up co-ordinates such that positive y is downwards from point of drop. The displacement of first body at $t = t$ after t_0 but before $\boxed{\text{reaching ground}}$ is

$$s_1 = \frac{1}{2}gt_0^2 + \frac{1}{2}g(t-t_0)^2 \tag{3.9}$$

While for second body is,

$$s_2 = ut + \frac{1}{2}g(t-t_0)^2 \tag{3.10}$$

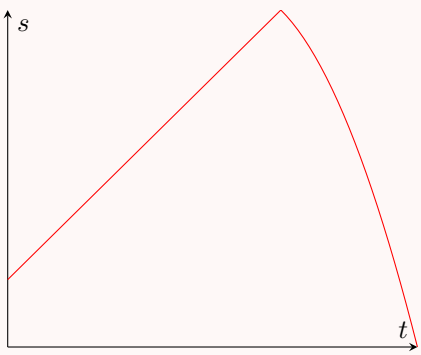
Thus,

$$s_{1,2} = \frac{1}{2}g(t_0)^2 + t(gt_0 - u) \tag{3.11}$$

Thus, before reaching ground, $s-t$ graph is linear. However, after first body reaches ground,

$$s_{1,2} = ut + \frac{1}{2}g(t)^2 \tag{3.12}$$

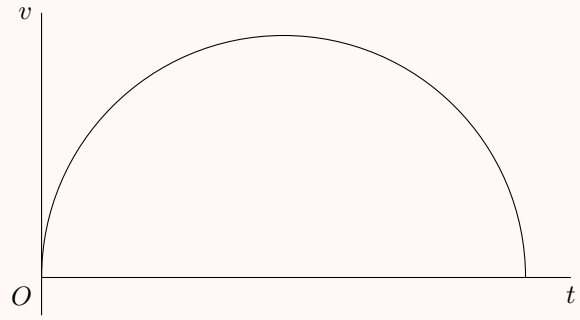
which is a parabola. Thus, overall graph is,



□

Example 3.5.6

Find the total displacement if the $v - t$ graph is as follows,



The idea behind this is simple, the graph here is **not** a semicircle. It is an ellipse, because of dimensional limitations! The equations of circle, $v^2 + t^2 = r^2$ doesn't work because we cannot add two dimensionally different quantities.

Therefore, the graph is an ellipse, and we conclude by computing the area of the ellipse. I haven't given the data, since I wanted to mention the idea.

§3.6 Motion in Polar Co-ordinates

The polar co-ordinate axes are the two-dimesional subset of the 3-d cylindrical co-ordinates. Any point in polar co-ordinates is depicted by a system of two unit vectors, $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$. However, each of these are $\boxed{\text{dependent on the position of the particle}}$ and maybe written explicitly as $\hat{\mathbf{r}}(\theta)$ and $\hat{\boldsymbol{\theta}}(\theta)$.

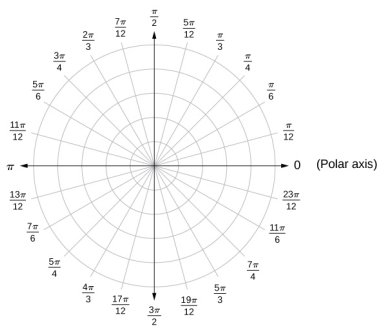


Figure 3.6: Polar axes

Here, $\hat{\mathbf{r}}$ points in the direction of increasing radius(along the straight lines in fig. 3.6) $\hat{\boldsymbol{\theta}}$ points in the direction of increasing θ (along the tangent to the circles). These two unit vectors are *orthogonal* at any point.

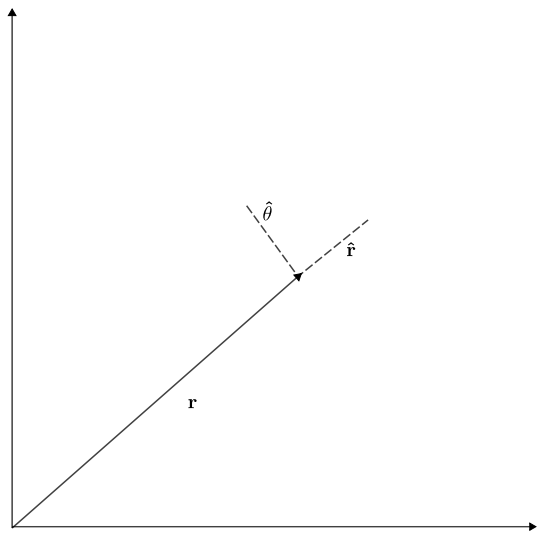


Figure 3.7: Unit vectors in polar co-ordinates

In fig. 3.7 if $\mathbf{r}$ makes an angle θ with the horizontal, then,

$$\hat{\mathbf{r}} = \cos\theta\hat{\mathbf{i}} + \sin\theta\hat{\mathbf{j}} \tag{3.13}$$

$$\hat{\boldsymbol{\theta}} = -\sin\theta\hat{\mathbf{i}} + \cos\theta\hat{\mathbf{j}} \tag{3.14}$$

§3.7 Kinematics in Polar Co-ordinates

Now let us formulate kinematics in polar co-ordinates. The position $\mathbf{r}$ can be written as

$$\mathbf{r} = r\hat{\mathbf{r}} \tag{3.15}$$

Velocity follows normally,

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + r\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} \tag{3.16}$$

or does it? What even is $\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t}$? Well the answer lies in the time-derivative of the unit vectors. Using the definitions in (3.13), we get,

$$\frac{\mathrm{d}\hat{\mathbf{r}}}{\mathrm{d}t} = \dot{\theta}\hat{\boldsymbol{\theta}} \tag{3.17}$$

$$\frac{\mathrm{d}\hat{\boldsymbol{\theta}}}{\mathrm{d}t} = -\dot{\theta}\hat{\mathbf{r}} \tag{3.18}$$

Using these results, we get that

$$\boxed{\begin{aligned} \mathbf{v} &= \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}} \qquad (3.19) \\ \mathbf{a} &= (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\boldsymbol{\theta}} \qquad (3.20) \end{aligned}}$$

The most interesting result are the various terms of acceleration in (3.20). First let us consider the terms along $\hat{\mathbf{r}}$. $\ddot{r}$ is the *radial* acceleration. It is the change is radial speed. The second term $-r\dot{\theta}^2$ is the *centripetal* acceleration. It arises from a change in tangential velocity.

Now let us consider the terms along $\hat{\boldsymbol{\theta}}$. The $r\ddot{\theta}$ term is the tangential acceleration resulting from change in tangential speed. The other term $2\dot{r}\dot{\theta}$ is the *coriolis* acceleration. I shall discuss it when discussing rotating frames.

§3.8 Free Fall

When an object is dropped downwards freely, it undergoes *free fall*. We ignore air resistance, until we're told not to and in general also talk about cases of the object being thrown instead of dropped. An object dropped from any height undergoes accelerations $\mathbf{a}$ where $\|\mathbf{a}\| = g$. The direction and sign of acceleration is entirely dependent on us.

In general, it is a good idea to setup our axes such that the direction above ground is positive and towards the ground is negative. The acceleration during free fall is always towards the ground. Therefore, for these cases $\mathbf{a} = -g\hat{\mathbf{j}}$.

Some important points of note are that "g" does not become positive or negative or whatever, this is a common misconception. Instead, we just decide where exactly the positive direction is and setup g according to that. When a ball is thrown with some velocity u say in the positive directions (change it into $-u$ if in the negative direction), then we have,

Algorithm 3.8.1 — The following are special cases of equations of motion.

1. $v = u - gt$
2. $v^2 = u^2 - 2gh$
3. $h = ut - \frac{1}{2}gt^2$

The case when a ball is thrown upwards is exactly symmetrical to this one, but there the ball goes up, reachs $v = 0$ and then goes down. Also note that all of these equations are vector equations.

Example 3.8.2

Two balls are thrown in intervals of 2 seconds with velocity u each. Find the time when they collide.

Solution. Note that when they collide, their displacement from ground must be the same! Setting the origin at ground, and direction of away from ground as positive, we have,

For the first ball, $h_1 = ut - \frac{1}{2}gt^2$. Since the second ball is thrown 2 seconds later, $h_2 = u(t-2) - \frac{1}{2}g(t-2)^2$. Now, we just have $h_1 = h_2$ and the calculation is trivial. □

The thing to note in this example is that you do not need to setup two cases, one in which g is negative and one in which g is positive since it is in the direction of velocity. Just note that these equations are vectorial and we are done.

Some quick formulas to remember because they help a lot,

Algorithm 3.8.3 — For an object dropped from height h,

1. Time to reach ground, $t = \sqrt{\frac{2h}{g}}$.
2. Velocity when it reaches ground $v = \sqrt{2gh}$.

Deriving them is easy, and they can be generalised as when it travels height h between the point of drop to anywhere, instead of ground. Clearly, both still follow there.

Also, always remember,

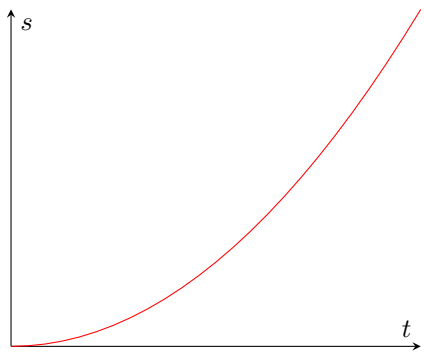
When an object is thrown/dropped from another object traveling with some velocity and acceleration, it inherits its velocity but not its acceleration.

add screw/lift example

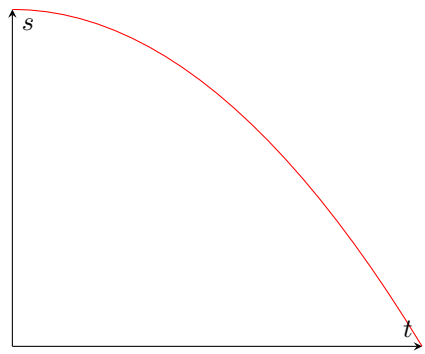
§3.9 Graphs in Motion

In the graphs we are going to discuss, acceleration will either be 0 or constant. These graphs help to quickly solve problems, especially related to total distance and displacement.

Let us consider the direction away from origin positive, and towards origin negative. Then, consider constant acceleration, $a = c$. We have the following displacement time graphs,



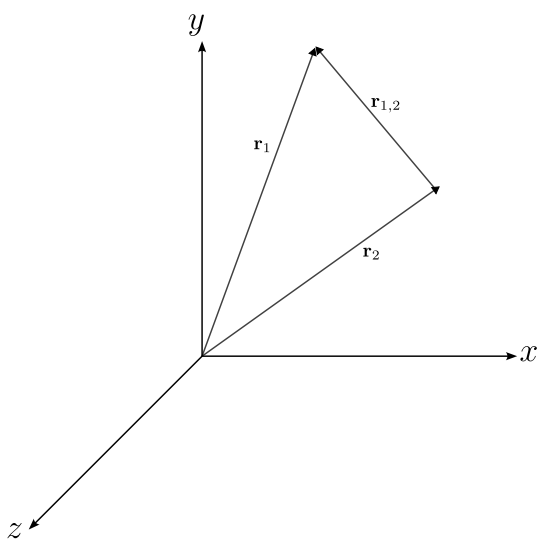
(a) For $c > 0$.



(b) For $c < 0$

§3.10 Relative Motion

There is a very neat technique to solve questions of kinematics, it is to switch our reference frames. Let the base reference frame be that of earth, then,



If $\mathbf{r}_1$ evolves with time, $r_1(t)$. And so does $r_2(t)$. Then,

$$\begin{aligned}
\mathbf{r}_1 &= \mathbf{r}_{1,2} + \mathbf{r}_2 \\
\dot{\mathbf{r}}_1 &= \dot{\mathbf{r}}_{1,2} + \dot{\mathbf{r}}_2 \\
\mathbf{v}_1 &= \mathbf{v}_{1,2} + \mathbf{v}_2 \\
\mathbf{v}_{1,2} &= \mathbf{v}_1 - \mathbf{v}_2
\end{aligned}$$

Obviously if the vectors face the same direction in the same plane, then, the components subtract. If they face opposite direction, then the components add.

$\mathbf{v}_{1,2}$ is called the velocity of *1* w.r.t *2*. When doing problems, try to setup a reference frame such that velocity of any single object vanishes. This allows for a very neat simplification of problems.

§3.11 Motion in higher dimensions

We can mostly safely extend our results from one dimension to higher dimensions. So for instances, velocity of a particle whose trajectory is defined as $x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ can simply be computed as,

$$v = \dot{x}\hat{\mathbf{i}} + \dot{y}\hat{\mathbf{j}} + \dot{z}\hat{\mathbf{k}}$$

Such a thing is not true for non-cartesian co-ordinates, as we say in the case of polar forms. Due to the orthogonality of the cartesian unit vectors, and using dot products we simply have,

$$v_x = \dot{x}, v_y = \dot{y}, v_z = \dot{z}$$

And the same is true for all other vectors. This gives us the power to resolve our vector components into components alongside unit vectors and consider independent one dimensional motion for each of them.

§3.12 Projectile Motion

Projectile motion is one of the most important motion to consider in 2 dimensions. It is the motion of a projected body. Generally, we only care about the motion of a body projected from ground under the influence of gravity.

Let a body be projected at an angle θ from the ground at velocity $\mathbf{v}$ along θ. Then, by resolving $\mathbf{v}$ into $v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}}$, we have,

$$x = v_x t\hat{\mathbf{i}}$$
$$y = v_y t - \frac{1}{2}gt^2\hat{\mathbf{j}}$$

Doing some manipulations, we may note that $v_x = v\sin(\theta)$, $v_y = v\cos(\theta)$. Also, substituting $t = x/v\sin(\theta)$, we have,

$$\boxed{y = x\tan(\theta) - \frac{gx^2}{2v^2\cos^2(\theta)}} \tag{3.21}$$

The path of the projectile is parabolic as shown in fig. 3.9.

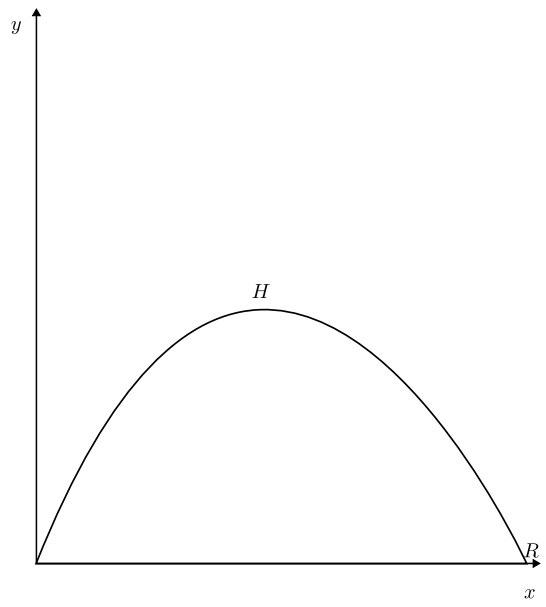


Figure 3.9: Trajectory of a projectile

Here, H is the maximum height of the particle. Clearly, when the projectile is at H, $v_y = 0$. We have, $v_y^2 = 2gH$,

$$\boxed{H = \frac{u^2 \sin^2(\theta)}{2g}} \tag{3.22}$$

The time period of the projectile can be simply calculated as well, since at the final time, T, $y = v_y t - 1/2gt^2 = 0$ and,

$$\boxed{T = \frac{2v \sin(\theta)}{g}} \tag{3.23}$$

R is the range the particle covers horizontally in time T and is clearly,

$$\boxed{R = \frac{v^2 \sin(2\theta)}{g}} \tag{3.24}$$

Note that if the particle is projected at $\theta = \pi/4$, the range is maximum and is $u^2/2g$. Otherwise, for $0 < \theta < \pi/2$, $\sin(2\theta)$ attains the same value for two θ_1, θ_2. These θ_1, θ_2 are complementary by trigonometry.

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PART C

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